WTW 158 Worksheet 7 on Differentiation
In order to be sure that you have mastered differentiation you need to complete the following:

A. The 200+ Challenge. This challenge consists of problems in the Study Guide under Technique Mastering, pages 44 - 48, sections I to K. The solutions are available on ClickUP. Complete this at home.

B. TEXTBOOK PROBLEMS
p204 no 47, 59, 61, 63
p215 no 25, 27, 39, 49, 51, 75
p223 no 23, 31, 33, 37, 39, 45, 47
p264 no 1, 3, 33, 35, 51

C If you have done all of the above and you need more challenging problems then you can try your hand at The Ultimate Differentiation Challenge, available on ClickUP under the tab Worksheets. This is not for everyone and you should only attempt it if you have truly mastered differentiation.

D. TEST QUESTIONS
Question 1
The equation of the tangent line to the graph of $y = e^{2x}$ at the point where $x = 0$ is

[a] $y = (2e^{2x})x + 1$	[b] $y = (2e^{2x})x$	[c] $y = 2x$	
[d] $y = x$	[e] $y = 2x + 1$	[f] $y = x + 1$	[g] None of these

Question 2
If $y^3 \sin x + \cos y = x$ then $\dfrac{dy}{dx} =$

[a] $\dfrac{1 - y^3 \cos x}{3y^2 \sin x - \sin y}$	[b] $\dfrac{1}{3y^2 \cos x + \cos y}$	[c] $\dfrac{1 - y^3 \cos x}{3y^2 \sin x}$
[d] None of these		

Question 3
Indien $f(x) = \cosh(e^{2x})$, dan is $f'(x) =$
If $f(x) = \cosh(e^{2x})$, then $f'(x) =$

[a] $\sinh(e^{2x})$	[b] $e^{2x}\sinh(e^{2x})$	[c] $-2e^{2x}\sinh(e^{2x})$	[d] $2e^{2x}\sinh(e^{2x})$
[e] None of these			

E. SELECTED ANSWERS
A. Answers on the 200+ Challenge is available on ClickUP under Technique Mastering
B. Answers in textbook.
D. 1 [e], 2 [a] 3 [d]

F. Start on Unit 4.1 p283 no 27, 33, 39, 49, 55, 61